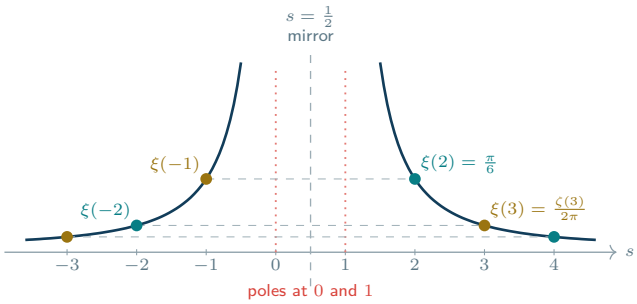
NUMBER THEORY ON ONE PAGE

# The completed zeta function at 2 and 3

Even arguments give powers of $\pi$, odd arguments give new constants, and $\Gamma$ decides which.

## 1 / THE RULE AND THE PICTURE

### One function with three components, mirrored about one half




teal: even argument, gold: odd argument

The completed zeta function has three components, a power of $\pi$, a Gamma function and $\zeta$ itself, and is symmetric about $s = \frac{1}{2}$:

$$\xi(s) = \pi^{-s/2}\,\Gamma\!\left(\tfrac{s}{2}\right)\zeta(s) = \xi(1-s)$$

It has poles at 0 and 1, and at $s = 2$ and $s = 3$ its values are $\xi(2) = \frac{\pi}{6}$ and $\xi(3) = \frac{\zeta(3)}{2\pi}$, repeated at $-1$ and $-2$ by the symmetry. Riemann's own $\xi$ carries an extra factor $\frac{1}{2}s(s-1)$ that removes the poles.

## 2 / THE COMPONENTS AT $s = 2, \ldots, 6$

### Even $s$ gives a power of $\pi$, odd $s$ keeps a zeta value

| $s$ | $\pi^{-s/2}$ | $\Gamma(s/2)$ as a factorial | $\zeta(s)$ | $\xi(s)$ | value |
|---|---|---|---|---|---|
| 2 | $\pi^{-1}$ | $\Gamma(1) = 0! = 1$ | $\frac{\pi^2}{6}$ | $\frac{\pi}{6}$ | 0.5236 |
| 3 | $\pi^{-3/2}$ | $\Gamma(\frac{3}{2}) = \frac{1}{2}! = \frac{\sqrt{\pi}}{2}$ | 1.2021 (Apéry) | $\frac{\zeta(3)}{2\pi}$ | 0.1913 |
| 4 | $\pi^{-2}$ | $\Gamma(2) = 1! = 1$ | $\frac{\pi^4}{90}$ | $\frac{\pi^2}{90}$ | 0.1097 |
| 5 | $\pi^{-5/2}$ | $\Gamma(\frac{5}{2}) = \frac{3}{4}\sqrt{\pi}$ | 1.0369 | $\frac{3\zeta(5)}{4\pi^2}$ | 0.0788 |
| 6 | $\pi^{-3}$ | $\Gamma(3) = 2! = 2$ | $\frac{\pi^6}{945}$ | $\frac{2\pi^3}{945}$ | 0.0656 |

Even rows have an ordinary factorial and give a rational multiple of $\pi^{s/2}$, and odd rows have a half-integer factorial carrying $\sqrt{\pi}$, which leaves $\zeta(s)$ over a power of $\pi$.

## 3 / WHY EVEN AND ODD DIFFER

### The Gamma function, Euler's Bernoulli values and the reflection each treat the two cases differently

First, $\Gamma(k) = (k-1)!$ while $\Gamma(k+\frac{1}{2}) = \frac{(2k)!\sqrt{\pi}}{4^k k!}$, so the $\sqrt{\pi}$ supplies the missing half power of $\pi$ and

$$\xi(2k) = \frac{(k-1)!}{\pi^k}\,\zeta(2k), \qquad \xi(2k+1) = \frac{(2k)!}{4^k k!}\,\frac{\zeta(2k+1)}{\pi^k}.$$

Second, Euler's $\zeta(2k) = \frac{(-1)^{k+1}B_{2k}(2\pi)^{2k}}{2(2k)!}$ with Bernoulli numbers makes $\xi(2k)$ a rational multiple of $\pi^k$, while no such formula is known for $\zeta(2k+1)$. Third, the reflection swaps parity: $\xi(2k) = \xi(1-2k)$, where $\zeta(1-2k) = -\frac{B_{2k}}{2k}$ is rational ($\zeta(-1) = -\frac{1}{12}$, $\zeta(-3) = \frac{1}{120}$), but $\xi(2k+1) = \xi(-2k)$, where $\zeta$ has a trivial zero cancelled by a pole of $\Gamma$, so the value survives only as a slope, $\zeta'(-2k) = \frac{(-1)^k(2k)!\,\zeta(2k+1)}{2(2\pi)^{2k}}$, for example $\zeta(3) = -4\pi^2\zeta'(-2)$.

## 4 / HOW IT DEVELOPED

### From the Basel sum to the odd values still unsolved

| year | who | result |
|---|---|---|
| 1735 to 1740 | Euler (after Mengoli, 1650) | finds $\zeta(2) = \frac{\pi^2}{6}$ from the product for $\sin \pi x$, then $\zeta(2k)$ through Bernoulli numbers |
| 1749 | Euler | conjectures the reflection relation between $\zeta(s)$ and $\zeta(1-s)$ |
| 1859 | Riemann | continues $\zeta$ to the complex plane, proves the functional equation from the theta function and introduces the completed function |
| 1882 | Lindemann | proves that $\pi$ is transcendental, so every $\zeta(2k)$ is transcendental |
| 1978 | Apéry | proves that $\zeta(3)$ is irrational |
| 2000 to 2001 | Rivoal, Ball, Zudilin | infinitely many odd values are irrational, and one of $\zeta(5), \zeta(7), \zeta(9), \zeta(11)$ is irrational |

No closed form for any odd value is known, and in the refereed literature the irrationality of $\zeta(5)$ is still open.

## 5 / WHERE THE COMPONENTS COME FROM

### The Gaussian supplies $\pi^{-s/2}\Gamma(s/2)$ and the sphere explains it

Since $\int_0^\infty e^{-\pi n^2 x} x^{s/2-1} dx = \pi^{-s/2}\Gamma(\frac{s}{2})n^{-s}$, summing over $n$ gives $\xi(s) = \int_0^\infty \psi(x)\, x^{s/2-1} dx$ with $\psi(x) = \sum_{n\geq 1} e^{-\pi n^2 x}$, and because the Gaussian is its own Fourier transform $\theta = 1 + 2\psi$ obeys $\theta(\frac{1}{x}) = \sqrt{x}\,\theta(x)$, which is the reflection $\xi(s) = \xi(1-s)$. Geometrically $\pi^{-s/2}\Gamma(\frac{s}{2}) = \frac{2}{S_{s-1}}$, where $S_{s-1} = \frac{2\pi^{s/2}}{\Gamma(s/2)}$ is the area of the unit sphere in $s$ dimensions ($2\pi, 4\pi, 2\pi^2, \frac{8\pi^2}{3}, \pi^3$ for $s = 2, \ldots, 6$), so $\xi(s) = \frac{2\zeta(s)}{S_{s-1}}$.

### ASIDE / TRY THESE

Check $\xi(5) = \frac{3\zeta(5)}{4\pi^2} = 0.0788 = \xi(-4)$ and $\zeta'(-4) = 0.00798$ from the slope formula with $k = 2$.

### LINKS TO OTHER SHEETS

The geometry sheet that follows uses $\xi(2)$, $\xi(3)$ and $\xi(4)$ in photon gases, and the dyadic sheet gives $\xi(2) = \frac{\pi}{6}$.

**Convention.** $\xi(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$; $B_n$ Bernoulli numbers; $S_{s-1}$ area of the unit sphere in $s$ dimensions; all identities and values were checked to many digits.

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